

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester III)**  
**Data Structures Practical**

## Practical - 3

|                                 |                                      |
|---------------------------------|--------------------------------------|
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| Class : SYIT                    | Batch : 01                           |
| Date of Assignment : 02-07-2026 | Date/Time of Submission : 02-07-2026 |

### Stack Application:

**Write a program to:**

**a. Implement a stack using an array.**

**Code:-**

```
import array as arr

stack1 = arr.array('i')

print(stack1)

def pop():
    print("Pop Operation")
    stack1.pop()
    print(stack1)

def push():
    print("push Operation")
    ele = int(input("Enter your element: "))
    stack1.append(ele)
    print(stack1)
```

```
def peek():
    print("peek Operation")
    print(stack1[-1])
    print("Panvir S031")|
```

```

def peek():
    print("peek Operation")
    if(isEmpty()==0):
        print("Stack is empty")
    else:
        print(stack1[-1])
        print("Panvir S031")

def isEmpty():
    print("isEmpty Operation")
    if len(stack1)==0:
        return 0

```

```

def pop():
    print("Pop Operation")
    if(isEmpty()==0):
        print("Stack Underflow")
    else:
        stack1.pop()
        print(stack1[-1])
        print("Panvir S031")

```

```

num = 0
while(num<5):
    print("1. Pop 2. push 3. peek 4. isempty 5. exit")
    num = int(input("select ch: "))

    if(num==1):
        pop()
    elif(num==2):
        push()
    elif(num==3):
        peek()
    elif(num==4):
        isEmpty()
    else:
        print("program closed")

print("Panvir S031")

```

### Output:-

```

1. Pop 2. push 3. peek 4. isempty 5. exit
select ch: 2
push Operation
Enter your element: 16
array('i', [13, 14, 15, 16])
Panvir S031

```

```
1. Pop 2. push 3. peek 4. isempty 5. exit
select ch: 1
Pop Operation
array('i', [13, 14, 15])
Panvir S031
```

```
1. Pop 2. push 3. peek 4. isempty 5. exit
select ch: 3
peek Operation
15
Panvir S031
```

```
array('i')
1. Pop 2. push 3. peek 4. isempty 5. exit
select ch: 1
Pop Operation
isEmpty Operation
Stack Underflow
1. Pop 2. push 3. peek 4. isempty 5. exit
select ch: 3
peek Operation
isEmpty Operation
Stack is empty
1. Pop 2. push 3. peek 4. isempty 5. exit
select ch: 4
isEmpty Operation
```

### Cod:-

```
def push():
    print("push Operation")
    if len(stack1)==si:
        print("stack overflow")
    else:
        ele = int(input("Enter your element: "))
        stack1.append(ele)
        print(stack1)
        print("Panvir S031")
```

### Output:-

```
1. Pop 2. push 3. peek 4. isempty 5. exit
select ch: 2
push Operation
Enter your element: 13
array('i', [11, 12, 13])
Panvir S031
1. Pop 2. push 3. peek 4. isempty 5. exit
select ch: 2
push Operation
stack overflow
```